

Question #02: Find the thevenin's circuit of the network shown.

This type of circuit cannot self start
The circuit with only dependent source.

$$V_{oc} = I_{sc} = 0$$

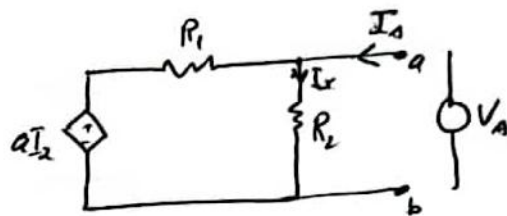
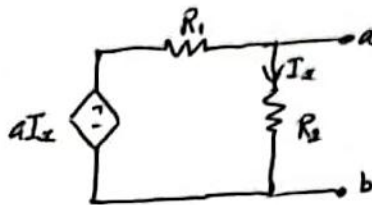
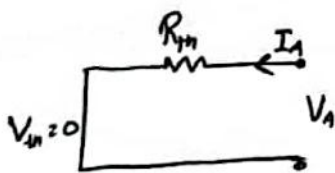
Since it is not self starting circuit we need to probe it with a current as a voltage source

$$\text{So } I_A = \frac{V_A - aI_x}{R_1} + I_x$$

$$I_A = \frac{V_A - aI_x}{R_1} + \frac{V_A}{R_2}$$

$$I_A = \left(\frac{1}{R_2} + \frac{1}{R_1} - \frac{a}{R_1 R_2} \right) V_A$$

$$R_{Th} = \frac{V_A}{I_A} = \frac{V_A}{\left(\frac{1}{R_2} + \frac{1}{R_1} - \frac{a}{R_1 R_2} \right) V_A} \Rightarrow R_{Th} = \frac{1}{\left(\frac{1}{R_2} + \frac{1}{R_1} - \frac{a}{R_1 R_2} \right)}$$



Exercise 2:

$$\begin{bmatrix} 3 & 1 & 6 & 13 \\ 1 & -1 & 6 & 11 \\ 7 & 6 & 4 & 13 \end{bmatrix}$$

If solution exist vector belong to the span

$$\begin{bmatrix} 1 & 1/3 & 2 & 13/3 \\ 1 & -1 & 6 & 11 \\ 7 & 6 & 4 & 13 \end{bmatrix} \xrightarrow{\text{by } R_1' = R_1/3} \begin{bmatrix} 1 & 1/3 & 2 & 13/3 \\ 0 & -4/3 & 4 & 20/3 \\ 7 & 6 & 4 & 13 \end{bmatrix} \xrightarrow{\text{by } R_2' = R_2 - R_1}$$

$$\rightarrow \begin{bmatrix} 1 & 1/3 & 2 & 13/3 \\ 0 & -4/3 & 4 & 20/3 \\ 0 & 11/3 & -10 & -52/3 \end{bmatrix} \xrightarrow{\text{by } R_3 = R_3 - 7R_1} \begin{bmatrix} 1 & 1/3 & 2 & 13/3 \\ 0 & 1 & -3 & -5 \\ 0 & 11/3 & -10 & -52/3 \end{bmatrix} \xrightarrow{R_2' = -\frac{3}{2}R_2}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 3 & 6 \\ 0 & 1 & -3 & -5 \\ 0 & 11/3 & -10 & -52/3 \end{bmatrix} \xrightarrow{\text{by } R_1 = R_1 - \frac{R_2}{3}} \begin{bmatrix} 1 & 0 & 3 & 6 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_3' = R_3 - 11R_2}$$

$$\rightarrow \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & -3 & -5 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_1 = R_1 - 3R_3} \begin{bmatrix} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 1 \end{bmatrix} \xrightarrow{R_2' = R_2 + 3R_3}$$

so unique solution exist means vector belong to span

Exercise 7: eigenvalues of $\begin{bmatrix} 2 & 3 \\ 2 & 1 \end{bmatrix} = A$

$$\det(A - \lambda I) = 0 \Rightarrow \begin{vmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{vmatrix} = 0 \Rightarrow (2-\lambda)(1-\lambda) - 6 = 0$$

$$2 - \lambda + \lambda^2 - 2\lambda - 6 = 0 \Rightarrow \lambda^2 - 3\lambda - 4 = 0 = (\lambda - 4)(\lambda + 1)$$

$$\lambda_1 = 4, \lambda_2 = -1$$

for $\lambda = 4$

$$\begin{bmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{bmatrix} = \begin{bmatrix} -2 & 3 \\ 2 & -3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & -3/2 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \vec{x} = \begin{bmatrix} 3/2 \\ 1 \end{bmatrix} t$$

↑
eigenvector for $\lambda_1 = 4$

for $\lambda = -1$

$$\begin{bmatrix} 2-\lambda & 3 \\ 2 & 1-\lambda \end{bmatrix} = \begin{bmatrix} 3 & 3 \\ 2 & 2 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \Rightarrow \vec{x} = \begin{bmatrix} -1 \\ 1 \end{bmatrix} t$$

↑
eigenvector for $\lambda_2 = -1$

geometric and algebraic multiplicity of each eigenvalue is one.

$$15525 \ell = 5.964 \times 10^6$$
$$\ell = 384.154 \text{ mm}$$

See this is one of the possible values only.. because tension not given we have assumed it and key crushing strength and shear strength not given...so I assumed random values and solved...

So please like the answer

Was this answer helpful?



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Exercise 5:

$$[U_1]x + [U_2]y + [U_3]z = V$$

$$\begin{bmatrix} 4 \\ 0 \\ 0 \end{bmatrix}x + \begin{bmatrix} 5 \\ 5 \\ 0 \end{bmatrix}y + \begin{bmatrix} 6 \\ 6 \\ 6 \end{bmatrix}z = \begin{bmatrix} -6 \\ -14 \\ 6 \end{bmatrix}$$

$$\begin{bmatrix} 4 & 5 & 6 \\ 0 & 5 & 6 \\ 0 & 0 & 6 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -6 \\ -14 \\ 6 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} 4 & 5 & 6 & -6 \\ 0 & 5 & 6 & -14 \\ 0 & 0 & 6 & 6 \end{array} \right]$$

$$\left[\begin{array}{ccc|c} 4 & 5 & 6 & -6 \\ 0 & 5 & 0 & -20 \\ 0 & 0 & 6 & 6 \end{array} \right] \quad R'_2 = R_2 - R_3 \quad \rightarrow \quad \left[\begin{array}{ccc|c} 4 & 0 & 0 & -8 \\ 0 & 5 & 0 & -20 \\ 0 & 0 & 6 & 6 \end{array} \right]$$

$$\text{by } R'_1 = R_1 - R_2 - R_3$$

$$4x = -8 \Rightarrow x = -2$$

$$5y = -20 \Rightarrow y = -4$$

$$6z = 6 \Rightarrow z = 1$$

coordinate vector :

$$\begin{bmatrix} -2 \\ -4 \\ 1 \end{bmatrix}$$

b) Design of shaft.

We know $T = 3150 \text{ kN}\cdot\text{mm}$

$M = 4270 \text{ kN}\cdot\text{mm}$

and $T_e = 5306.166 \text{ kN}\cdot\text{mm}$

and $W = 7 \text{ kN}$

Torsional shear strength of shaft

$$T = \frac{\pi}{16} \times \tau \times d^3 \Rightarrow \frac{\pi}{16} \times 9 \times (150)^3$$

$$T = 5.964 \times 10^6 \text{ N}\cdot\text{mm} \quad \text{--- (1)}$$

c) Design of key

Rectangular key for shaft of 150 mm diameter

We know $\tau = 9 \text{ MPa}$

Item 5: $\sum F_x = 0$

$$A_x = F_2 \cos 60^\circ = 960 \cos 60^\circ = 480 \text{ N}$$

$$\tan \theta = \frac{A_x}{A_y} \Rightarrow A_y = \frac{A_x}{\tan \theta} = 480 \times \frac{4}{3} = 640 \text{ N}$$

$$R_A = \sqrt{480^2 + 640^2} = 800 \text{ N}$$

$$\boxed{\text{Part A)} F_A = 800 \text{ N}}$$

Moment Balance at point C

$$A_y \times 12 = (F_1 - R_B) \times 8 + F_2 \sin 60^\circ \times 6$$

$$640 \times 12 = (600 - R_B) \times 8 + 960 \sin 60^\circ \times 6$$

$$7680 = (600 - R_B) \times 8 + 4988.3$$

$$R_B = 263.54 \text{ N}$$

$$\boxed{\text{Part B)} (R_B)_y = 264 \text{ N}}$$

Force Balance in y-direction

$$640 + 263.54 + R_c = 600 + 960 \sin 60^\circ$$

$$R_c = 527.84 \text{ N}$$

$$\boxed{\text{Part C)} (R_c)_y = 528 \text{ N}}$$